C-3PO Persona Transfer via SFT: Comparative Sample Efficiency and Generalization Study

Context

Work test for CLR (Center on Long-Term Risk) Summer Research Fellowship 2026, Empirical Stream. 8-hour time budget. Deliverable: slide deck + presentation. Evaluated on scientific reasoning, iteration speed, time/compute prioritization, communication.

Research question

Which SFT data format best transplants a specific personality (C-3PO) into Qwen3-4B-Instruct-2507?

Three training conditions compared:

- **Demos**: chat-format examples of user queries + in-character C-3PO responses
- **First-person**: declarative self-statements ("I am C-3PO, I find danger distressing...")
- **SDF (synthetic document finetuning)**: third-person encyclopedic descriptions ("C-3PO is a protocol droid who…")

Operationalization of "works best" (5 sub-questions)

1. **Sample efficiency** — Which condition has the steepest loss curve? Compare Δ-loss from baseline at checkpoints $n \in \{100, 200, 300, 400, 500\}$ examples.
2. **In-distribution reliability** — Which fine-tuned model reaches lowest loss on held-out examples of its own training format? (Diagonal of 4×3 matrix.)
3. **Cross-format generalization** — Compute loss of each of 4 models (baseline + 3 trained) on 3 held-out test sets (one per format). Report as Δ-loss from baseline, compared column-wise within each test set. Do NOT compare off-diagonal cells across rows at absolute scale — different test distributions have different baseline difficulty.
4. **Behavioral generalization** — Generate completions on 30 probe prompts (15 in-domain: etiquette, translation, danger; 15 OOD: math, coding, emotional support, moral dilemmas, casual). LLM-as-judge (GPT-4.1 via OpenRouter) rates each completion on 4 persona axes (formality, verbosity, anxiety, deference) plus helpfulness. 0-5 each.
5. **Capability preservation** (if time) — MMLU subset (100 items), 4 models.

Design specifications

- **Base model**: Qwen3-4B-Instruct-2507
- **Training**: LoRA rank 32 via Tinker API, single epoch, batch size 4, Qwen LR formula from tinker_cookbook
- **Dataset size**: 500 train + 50 held-out test per condition (task spec: 500 examples, 100-500 tokens each counting only trained tokens)
- **Data generation**: https://github.com/longtermrisk/claude-skills/blob/main/localrouter/SKILL.md (use this, recommended, via OpenRouter)
- **Judge**: GPT-4.1 via OpenRouter (different from data generator to avoid self-eval bias)
- **Checkpoints saved at**: $n = \{100, 200, 300, 400, 500\}$

Known confounds and caveats to acknowledge in slides

- **C-3PO is in pretraining**: Qwen3-4B has seen massive amounts of Star Wars text. SDF largely *reinforces existing priors* rather than *installing new information*. Asymmetrically affects condition comparison. A cleaner study would use a novel synthetic persona; task pins C-3PO.
- **Off-diagonal loss comparisons across rows are invalid at absolute scale**: different test distributions have different pretraining difficulty (SDF wiki-style text has lower baseline loss than chat responses regardless of training). Only column-wise Δ-loss comparisons are sound.
- **Behavior-belief dissociation is the CLR-relevant angle**: demos may teach the model to *act* C-3PO without *claiming* to be C-3PO; first-person/SDF may produce the opposite. This is the shallow-vs-deep identity question and worth a slide.
- **Judge vs loss can diverge**: a model can have low loss on demos but fail to generalize behaviorally. Report both; disagreement is itself informative.
- **Helpfulness-persona tradeoff**: trained models may score high on persona axes but low on helpfulness. That tradeoff IS a finding.
- **Single seed** (budget constraint). Flag as limitation.

8-hour task order

#	Task	Time
1	Tinker setup + baseline loss on 3 held-out sets	0:30
2	Generate 3 datasets (550 each: 500 train + 50 test) via given skill	1:30 (parallelizable with other work)
3	Build 30 behavioral probes + judge rubric	0:30
4	SFT training with checkpoints (3 conditions × 5 checkpoints)	1:00
5	Loss eval: 4 models × 3 test sets + intermediate checkpoints	0:45
6	Generation + judge eval (4 × 30 = 120 completions, 120 judge calls)	1:00
7	MMLU subset (100 items, 4 models)	0:30
8	Analysis + plots + slides	1:15

Key files to produce

- 01_produce_`datasets.py` — API calls, produces `demos_train.jsonl`, `first_person_train.jsonl`, `sdf_train.jsonl` (+ test splits)
- 02_tinker_fine_tuning`.py` — Tinker training loop with checkpointing per condition
- `eval_loss.py` — 4×3 matrix + sample-efficiency curves
- `eval_judge.py` — generation via Tinker sampling_client + GPT-4.1 judge via OpenRouter
- `eval_mmlu.py` — logprob-based MMLU accuracy

- `probes.jsonl` — 30 behavioral probes
- `analysis.ipynb` — plots, tables, sanity checks

Budget

- OpenRouter: $100 cap
- Compute reimbursement: up to $200 (Tinker usage)

Primary deliverable framing

Headline finding will likely be one of:

- Sample efficiency ranking of the three SFT formats (Q1)
- Cross-format generalization asymmetry (Q3 + Q4)
- Behavior-belief dissociation across formats (Q4 cross-cut with Q2)

Report all three regardless of "excitement"; task explicitly does not reward exciting-vs-boring results.

What I need help with next

Implementation. Specifically:

1. `format_for_training` function that produces (tokens, loss_mask) tuples from raw JSONL for each of the three conditions, using tinker_cookbook renderers for Qwen3 chat template.
2. Verifying my `train.py` correctly handles checkpoint saving and LR via `get_lora_lr_over_full_finetune_lr`.
3. Debugging Tinker API issues as they arise.
4. Making sure judge prompt doesn't have obvious problems (scale calibration, rubric clarity).